

cell membrane & transport

Biological membranes:

- all cells & organelles are composed of a **phospholipid bilayer**
- cell membranes provide a partially permeable membrane, are the site of chemical reactions and have a role in cell communication

Fluid mosaic model:

- fluid ⇒ phospholipids & protein molecules are able to move about & diffuse sideways within its monolayer
- mosaic ⇒ proteins scattered within membrane

Components of the plasma membrane:

1) phospholipids thickness of layer: 7nm

- composed of 1 glycerol, 2 fatty acids & 1 phosphate group
- hydrophilic phosphate head is polar, forms 'H' bonds with water, and stabilises the membrane
- hydrophobic fatty acid tail is non-polar & repels water

- * barrier to water-soluble substances
- * allow lipid-soluble, small, uncharged molecules to diffuse through bilayer
- * help maintain fluidity of membrane

what affects membrane fluidity →

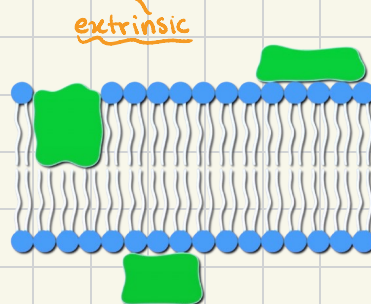
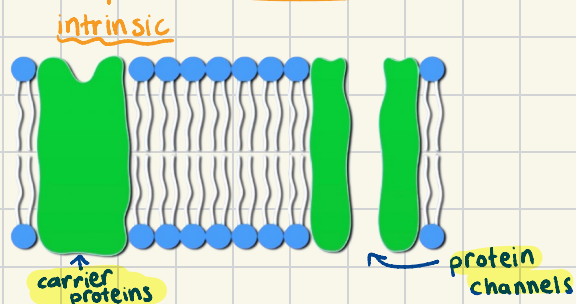
- high temp. = high KE = more fluid
 - higher 'unsat:sat' ratio = more fluid
 - shorter phospholipid tails = more fluid
- + cholesterol

2) cholesterol

- small molecule not found in prokaryotes' cell membrane
- has hydrophilic head & hydrophobic head and fits between phospholipid molecules
- regulates fluidity of membrane, stabilises membrane, and blocks passage of very small ions
(especially hydrophobic layer)
- * low fluidity = cholesterol increases fluidity by preventing close packing of the phospholipid tails
- * at high fluidity = cholesterol decreases fluidity by reducing the mobility of the phospholipids

3) membrane proteins

- two types: intrinsic & extrinsic



→ fully embedded in bilayer, spanning from one side to the other

→ have hydrophobic amino acids on their outer surface which interacts with the hydrophobic fatty acid tails

* **PROTEIN CHANNELS** are channels lined with hydrophilic amino acids to allow water-soluble molecules/ions to diffuse through

* **CARRIER PROTEINS** can change shape/position to transfer molecules/ions across the membrane through diffusion or active transport

→ Located on either side of the membrane

→ Sometimes are attached to intrinsic proteins

* serve structural roles within membrane, act as enzymes & function as receptors for hormones & help with cell-to-cell adhesion

* also make glycoproteins & glycolipids

• 'glyco' means carbohydrate chain

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4) glycolipids

→ carbohydrate chains attached to phospholipids

→ act as antigens for cell-to-cell recognition & help in cell-to-cell adhesion

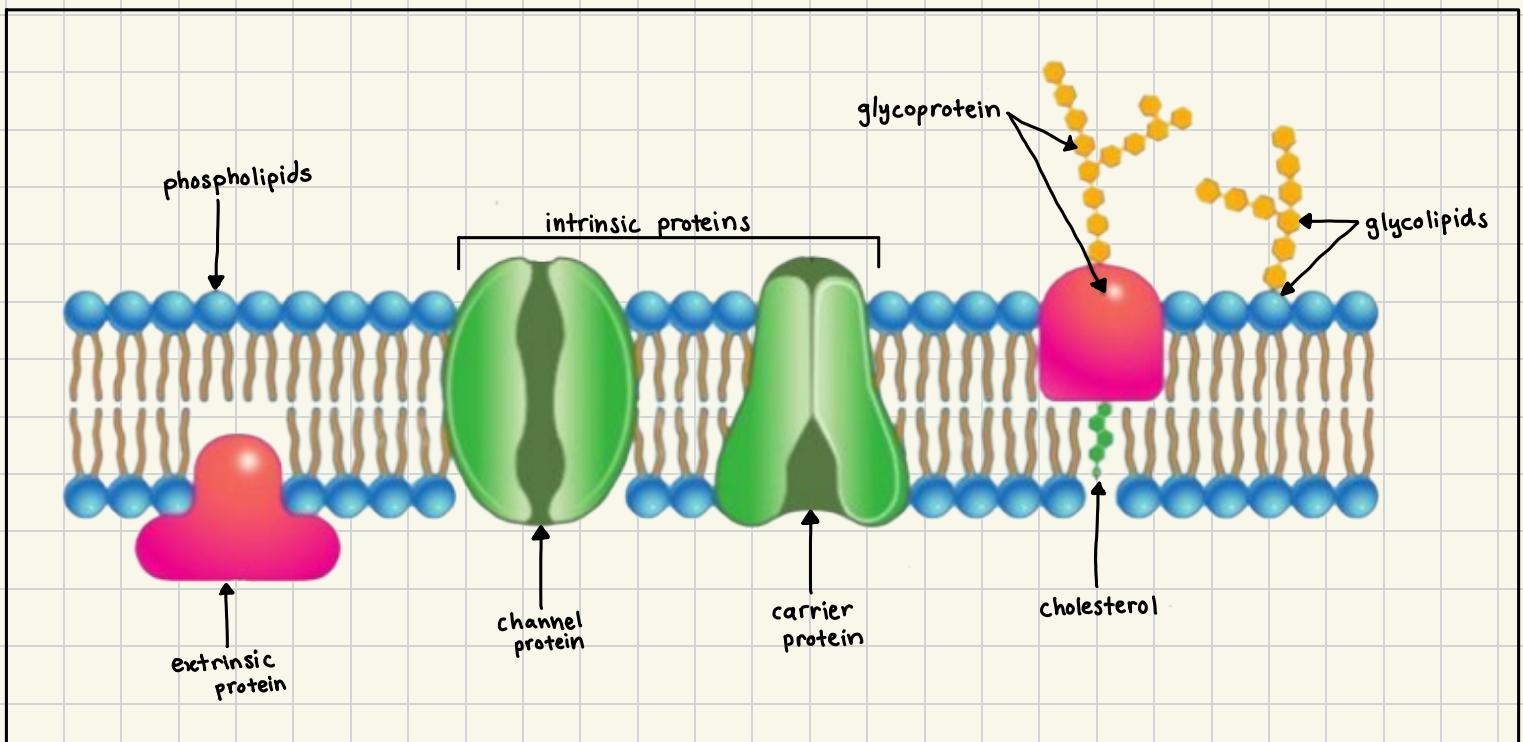
→ help maintain stability of the membrane

5) glycoproteins

→ carbohydrate chains attached to protein

→ act as antigens for cell-to-cell recognition & help in cell-to-cell adhesion

→ act as recognition sites for hormones & neurotransmitters

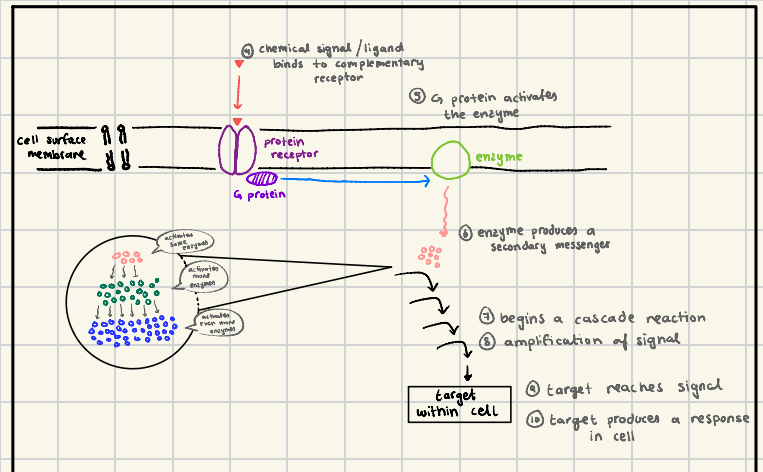


Cell signalling :

→ how cells detect & respond to stimuli ; how cells communicate

- 1) chemical signals / ligands are secreted from cells
- 2) ligands transported via bloodstream to target cells
- 3) ligands bind to specific & complementary cell surface receptors on the target cells
protein carriers

* signal passed into cell → transduction



→ responses can include increase in transcription of gene, movement, change in cell shape, secretion, activation of an enzyme, altered metabolism, opening an ion channel etc.

Types of ligands :

→ water-soluble ligands e.g. adrenaline, glucagon

↳ cannot pass through membrane, but recognized by receptor at plasma membrane

→ lipid-soluble ligands e.g. steroids, oestrogen

↳ can directly diffuse across the cell surface membrane

↳ binds to intracellular receptors in cytoplasm or nucleus

Mechanisms of transport across membranes :

① simple diffusion passive transport

→ the random net movement of molecules from a region of high conc. to a region of low conc., down the concentration gradient until equilibrium

→ in cells, this occurs in the phospholipid bilayer & no ATP is used

✓ non-polar, lipid-soluble, uncharged, small molecules (and water) can pass through bilayer

✗ polar, water-soluble, charged molecules cannot pass through the hydrophobic core of the bilayer

factors affecting the rate of simple diffusion →

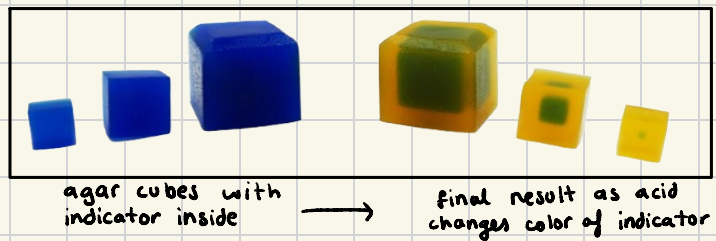
- higher conc. gradient / temperature = higher rate
- smaller molecules = higher rate
- higher $\frac{\text{surface area}}{\text{volume}}$ = higher rate

* surface area to volume ratios decrease with increasing size

	1	2	3
Length	1cm	2cm	3cm
SA	6cm ²	24cm ²	54cm ²
V	1cm ³	8cm ³	27cm ³
SA:V	6:1	3:1	2:1

*this can be shown through an experiment with agar blocks

- the agar cubes are impregnated with indicator
- they are put into petri dishes with an acid
- the depth of color change in each cube shows the diffusion of acid into the cubes



② facilitated diffusion [passive transport]

- diffusion through membrane transport proteins, down the conc. gradient
- involves channel proteins & carrier proteins so ions and polar molecules can pass through
- no ATP is used

✓ large, water-soluble, polar molecules can pass through the transport proteins

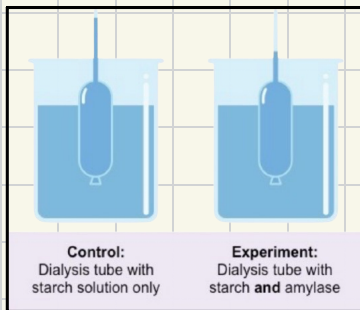
factors affecting the rate of facilitated diffusion ⇒

- higher conc. gradient / temperature = higher rate
- higher number of transport proteins available = higher rate
- higher surface area of membrane = higher rate

③ osmosis [passive transport]

- diffusion of water from a region of higher water potential (ψ) to low water potential across a partially permeable membrane until equilibrium & no ATP is used

VISKING TUBE EXPERIMENT

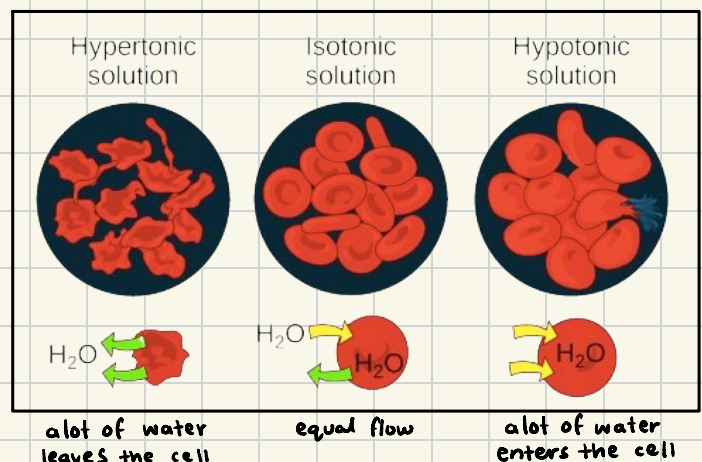


⇒ in control: starch cannot diffuse out because it's too big, but water diffuses into tube via osmosis resulting in raised water level. Benedict's solution test on solution in beaker would remain blue (negative)

⇒ in experiment: starch is broken down into maltose by amylase, & the maltose diffuses out, thus less water diffuses out by osmosis. Benedict's solution test on solution in beaker would turn brick-red (positive)

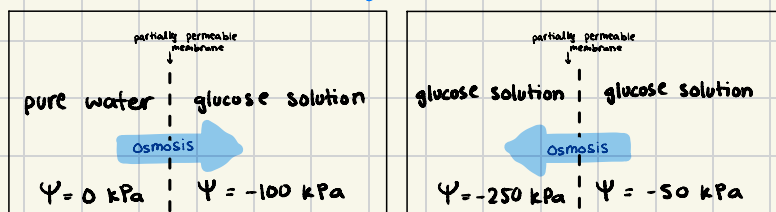
TYPES OF SOLUTION

- hypertonic is when the water potential of a solution is more negative than the cell
- isotonic is when the water potential of a solution is the same in the solution as the cell within the solution
- hypotonic is when the water potential of a solution is more positive than the cell

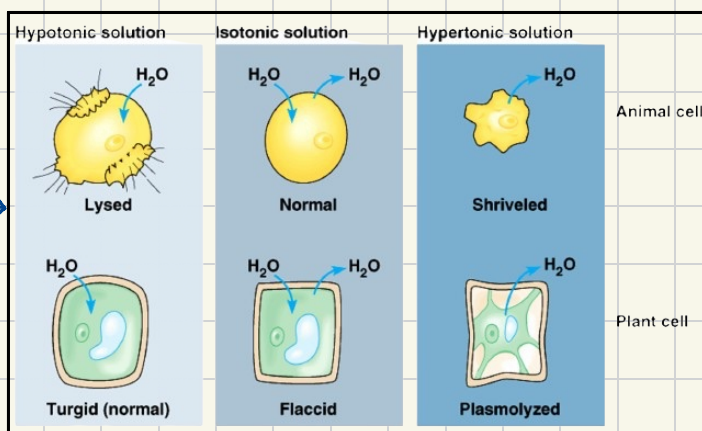


- ⇒ the symbol for water potential is Ψ and its calculated in kPa
- ⇒ water potential is always negative and the greatest value it can be is zero, which is the water potential of pure water

e.g



effect of osmosis on
plant & animal cells



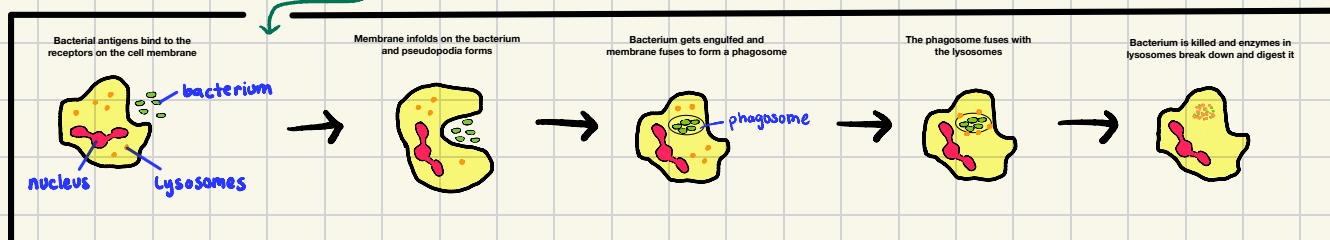
④ active transport

- movement of molecules/ions through carrier proteins from a region of low conc. to high conc., against the concentration gradient, using energy in the form of ATP which is needed for conformational change of carrier protein & results in steeper conc. gradient

Endocytosis & Exocytosis are both mechanisms to transport large quantities of substances and they require energy in the form of ATP

⑤ endocytosis

- transports substances into the cell
- solids through phagocytosis & liquids through pinocytosis



⑥ exocytosis

- transports substances out of cell
- substances packaged within vesicle with cell membrane to release the contents